

Hybrid Battery–Supercapacitor Solar Inverter with Smart Cooling and Fault Protection

1. Introduction / Abstract

In modern renewable energy systems, inverters are essential for converting DC power from solar panels or batteries into usable AC power. However, traditional inverters face several limitations — slow response during sudden load changes, reduced battery life, overheating under continuous operation, and lack of intelligent fault protection.

This project proposes the design and implementation of a **Hybrid Battery–Supercapacitor Solar Inverter** that overcomes these issues.

The system will integrate **a battery** for steady-state energy storage and **a supercapacitor** for handling short-term high current surges, resulting in improved efficiency and faster transient response.

The inverter will also include:

- An **MPPT (Maximum Power Point Tracking)** solar charge controller for maximum solar utilization.
- A **temperature-controlled cooling system** that adjusts fan speed based on real-time heat levels.
- A **fault detection and protection unit** to prevent damage from overload, short-circuit, or over-temperature conditions.

This combination makes the inverter **intelligent, efficient, and robust**, suitable for standalone or hybrid solar systems.

2. Objectives

1. To design and build a **DC–AC inverter** that can efficiently convert solar and battery energy to AC power.
2. To integrate a **hybrid energy storage system** (battery + supercapacitor) for improved performance under varying loads.
3. To develop an **MPPT charge controller** for optimal solar power utilization.

4. To implement a **smart cooling system** that automatically controls fan speed based on inverter temperature.
 5. To design a **hardware-based protection mechanism** for overcurrent, overtemperature, and short-circuit conditions.
 6. To display key parameters (voltage, current, temperature, load status, fault message) on an LCD or IoT dashboard.
-

3. Problem Statement

Conventional standalone inverters are limited by:

- Poor transient response to load changes
- Battery degradation due to high surge currents
- Inefficient cooling and energy losses
- Lack of integrated solar MPPT and smart protection features

This project aims to overcome these limitations by developing a **hybrid energy, self-protecting, and intelligent cooling inverter system**.

4. Methodology / Block Diagram Explanation

Main Functional Blocks:

1. **Solar Panel Input (DC Source)**
 - Provides renewable DC power.
 - Controlled via an MPPT algorithm for maximum power extraction.
2. **MPPT Controller (DC–DC Boost Converter)**
 - Steps up solar voltage and tracks the maximum power point dynamically.
 - Supplies DC power to the DC bus and charges the hybrid storage system.
3. **Hybrid Storage Unit (Battery + Supercapacitor)**
 - Battery handles continuous energy supply.
 - Supercapacitor provides high current during transient or sudden load demand.

- DC bus capacitor maintains system stability.

4. DC–AC Inverter Stage (H-Bridge)

- Converts DC to AC using PWM control (SPWM or unipolar PWM).
- Built with MOSFETs or IGBTs.

5. Smart Cooling System

- LM35/DS18B20 temperature sensor monitors inverter heat sink temperature.
- Microcontroller adjusts fan speed via PWM.

6. Protection Unit

- Uses current sensors (ACS712) and temperature sensors to detect faults.
- Triggers relays or MOSFET cut-off to isolate faulty sections.

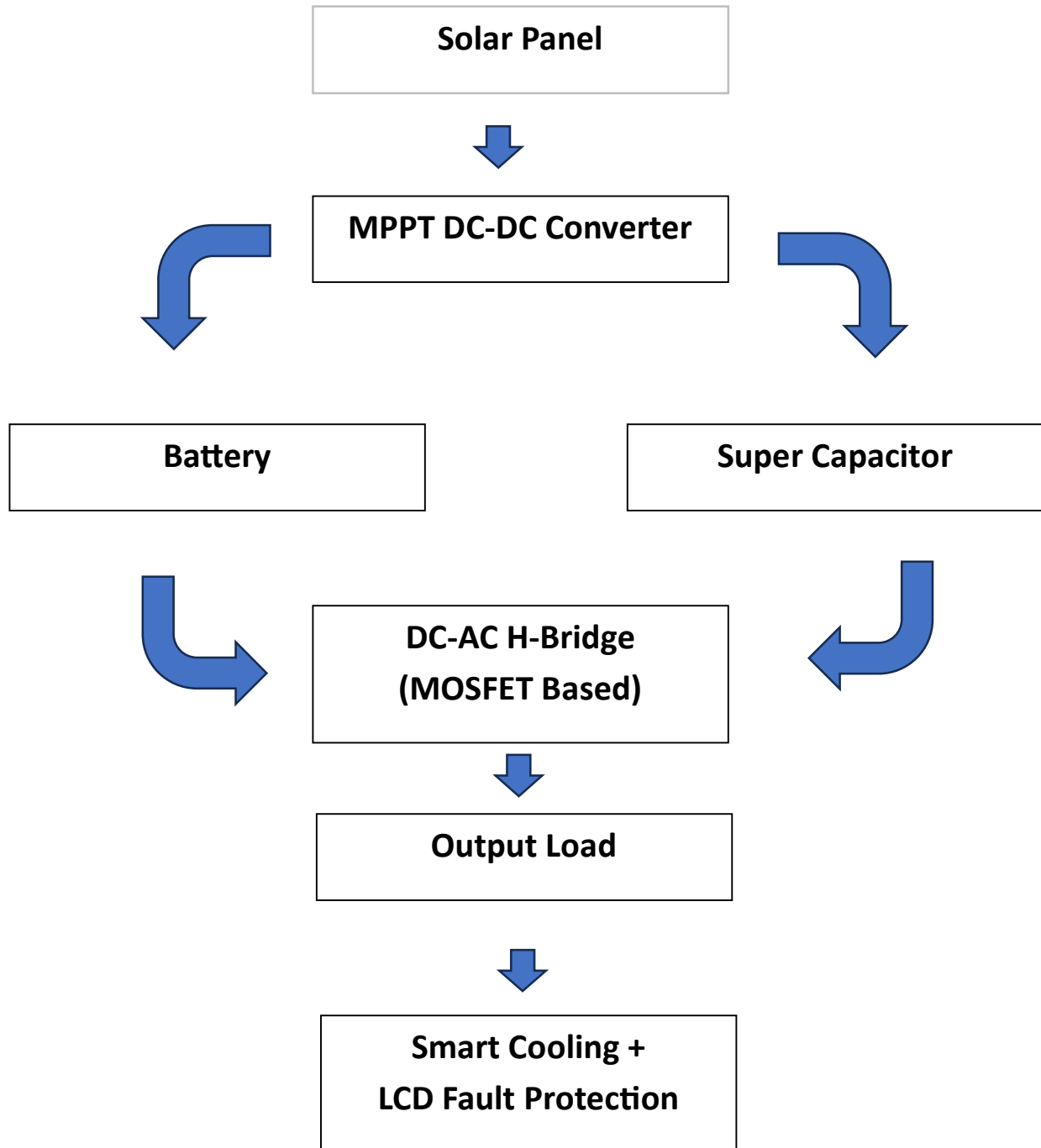
7. Control Unit (Microcontroller)

- Arduino/STM32 for overall control, sensing, and display management.
- Generates PWM for inverter and MPPT.

8. Output and Display Section

- 16x2 LCD or IoT dashboard shows voltage, current, temperature, and fault messages.
-

Block Diagram (Conceptual)



5. Hardware Components Required

Component	Description
Solar panel	DC input source (12V–24V)
MPPT module (custom)	Boost converter with feedback
Battery	12V Li-ion or Lead-acid
Supercapacitor bank	2.7V–5V per cell, 10–50F
MOSFETs (IRF540N / IRFZ44N)	Inverter bridge switches
Microcontroller	Arduino UNO / STM32
Driver IC	IR2110 or opto-isolated driver
Temperature sensor	LM35 / DS18B20
Current sensor	ACS712 / INA219
Relay module	For output and fault protection
Cooling fan	12V DC fan with PWM control
LCD Display	16x2 or 20x4
Transformer	12V-220V step-up (for AC output)
Capacitors, resistors, diodes	Passive components for filtering and smoothing

6. Expected Results

The inverter will:

- Efficiently convert DC to AC with pure sine PWM output.
- Maintain stable AC voltage even under sudden load changes.
- Increase battery lifespan by sharing surge currents with a supercapacitor.
- Automatically adjust cooling based on internal temperature.
- Detect and protect against overcurrent, short circuit, and overtemperature.
- Display system parameters and fault status in real time.

7. Applications

- Standalone solar energy systems (home, remote area)
 - Portable renewable power systems
 - Smart hybrid storage labs and EV applications
 - Educational and research projects on energy management
-

8. Novelty / Uniqueness

Feature	Traditional Inverter	Proposed Inverter
Energy storage	Battery only	Battery + Supercapacitor
Solar interface	External MPPT	Built-in MPPT
Cooling	Fixed fan	Smart temperature-controlled fan
Fault protection	Basic trip-off	Intelligent isolation + message
Monitoring	LED indicators	LCD + IoT/Display
Efficiency	Moderate	High (due to hybrid storage + MPPT)

9. Expected Outcome

By the end of the project, you will have a **fully functional prototype** of a **Hybrid Smart Solar Inverter** that:

- Operates efficiently and safely under varying loads.
 - Demonstrates modern renewable energy integration concepts.
 - Is more innovative than existing market designs.
 - Can be published in an IEEE or IEE conference as a new inverter control architecture.
-

10. References

1) A Survey of Battery–Supercapacitor Hybrid Energy Storage — Dong et al., *MDPI* (2022)

Why read: comprehensive survey of HESS design choices, modeling, sizing, and energy management strategies for battery + supercapacitor systems — excellent for understanding architectures, trade-offs, and state of the art.

How it helps your FYP: use it to choose topology, sizing method, and EMS strategies for sharing surge currents between supercaps and battery.

2) Supercapacitor Assisted Hybrid PV System for Efficient Energy Management — Piyumal et al., *MDPI Electronics* (2021)

Why read: practical design & experimental implementation of a PV system assisted by a supercapacitor with MPPT and charge control — includes control modes and performance data.

How it helps your FYP: gives a tested MPPT + supercap integration pattern you can adapt for your hybrid DC bus and transient handling.

3) Performance Evaluation of Hybrid Battery–Supercapacitor HESS (sizing & EMS) — ElGhanam et al., *Sustainability* / *MDPI* (2023)

Why read: presents optimal sizing and hierarchical energy management (H-HESS) for real systems and quantifies benefits (battery throughput reduction, improved lifetime).

How it helps your FYP: use its sizing and EMS results to justify your supercap/battery sizing and to demonstrate expected battery life improvement.

4) A Thermal Management Strategy for Inverter System Based on Predictive Control — (International Journal of Heat & Technology, 2023)

Why read: proposes a predictive-control based thermal management for inverter modules; includes modeling and energy-saving results from proactive thermal control.

How it helps your FYP: forms the theoretical basis for your smart cooling block (temperature prediction → PWM fan control) and gives a measurable metric to claim reduced cooling energy.

5) A Thorough Review of Cooling Concepts and Thermal Management for Inverters — Abramushkina et al., *Energies* (2021)

Why read: extensive review of cooling approaches (air, liquid, pin-fin, heat pipes) especially for modern WBG devices — good to select a practical cooling approach for your prototype.
How it helps your FYP: helps you pick an appropriate heat sink/fan solution and justify why a predictive PWM cooling strategy is better than on/off fans.

6) A Fault-Tolerant Topology for Single-Phase H-Bridge Inverters — Djaghloul et al. (2024/2025)

Why read: proposes an inverter topology and control that sustains operation under open/short failures — directly relevant to your fault protection / self-isolation feature.
How it helps your FYP: adapt concepts from this paper to design your protection unit and fault-isolation logic so the inverter can isolate faults without full shutdown.